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Unit – 5 ⇝ Inner Product Space (Tutorial Manual)

Method 1 ⇝ Real Inner Product Space

Examples of Method-1: Real Inner Product Space

A	1	Let $u = (u_1, u_2), v = (v_1, v_2) \in \mathbb{R}^2$. Check whether $\langle u, v\rangle$ defined as $\langle u, v\rangle = 4u_1v_1 + 6u_2v_2$ is an inner product on $\mathbb{R}^2$? **Answer: Yes**
A	2	Let $V = \mathbb{R}^2$ and $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ be vectors in V. Let k be a fixed positive number and define the function $\langle\ ,\ \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ by $\langle u, v\rangle = u_1v_1 + ku_2v_2$. Show that V is an inner product space.
A	3	Determine which of the following is inner product in $\mathbb{R}^3$. (1). $\langle u, v\rangle = 2u_1v_1 + u_2v_2 + 4u_3v_3$ (2). $\langle u, v\rangle = u_1v_1 - u_2v_2 + u_3v_3$ **Answer: (1). Yes, (2). No**
A	4	Let $V = \mathbb{R}^2$ and $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ be vectors in V. If $\langle u, v\rangle = u_1^2v_1^2 + u_2^2v_2^2$ then check that $(\mathbb{R}^2, \langle\ ,\rangle)$ is inner product space or not. **Answer: No**
B	5	Show that $\langle x, y\rangle = x_1y_1 - x_1y_2 - x_2y_1 + 2x_2y_2$ on $\mathbb{R}^2$ is an inner product.
B	6	Define inner product space. Let $u = (u_1, u_2)$ and $v = (v_1, v_2) \in \mathbb{R}^2$. Let $\langle u, v\rangle = 4u_1v_1 + u_1v_2 + u_2v_1 + 4u_2v_2$. Prove that $(\mathbb{R}^2, \langle\ ,\rangle)$ is an inner product space.
A	7	Let $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$ be two matrices in real matrix space $\mathbb{M}_{22}$ and let $\langle A, B\rangle = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4$. Show that $\langle A, B\rangle$ is an inner product in $\mathbb{M}_{22}$.

B	8	Let $\langle\ ,\ \rangle : \mathbb{M}_{23} \times \mathbb{M}_{23} \to \mathbb{R}, \langle A, B\rangle = \text{tr}(B^T A)$. Show that $(\mathbb{M}_{23}, \langle\ ,\ \rangle)$ is inner product space. Find $\langle A, B\rangle$ if $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. **Answer: 119**
C	9	Let $V = M_n(\mathbb{R})$ be the vector space of all $n \times n$ matrices with real entries. Prove first three conditions of inner product for the function defined by $\langle A, B\rangle = \text{tr}(A^T SB)$. Where S is fix symmetric matrix of order $n \times n$.
A	10	Let $p = a_0 + a_1x + a_2x^2$ and $q = b_0 + b_1x + b_2x^2$. Define the function $\langle\ ,\ \rangle : \mathbb{P}_2 \times \mathbb{P}_2 \to \mathbb{R}$ by $\langle p, q\rangle = a_0b_0 + a_1b_1 + a_2b_2$. Prove that $\mathbb{P}_2$ is inner product space.
B	11	Let $V = \mathbb{P}_2$ with inner product defined by $\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1)$. Show that $(\mathbb{P}_2, \langle\ ,\ \rangle)$ is inner product space. Let $p(x) = 3x^2 - 2$ and $q(x) = x$, find $\langle p, q\rangle$. **Answer: 0**
B	12	Let $V = \mathbb{P}_2$ with inner product defined by $\langle p, q\rangle = \int_0^1 p(x)q(x)\,dx$. Let $p(x) = 1 - x^2$ and $q(x) = 1 - x + 2x^2$, find $\langle p, q\rangle$ and verify $\langle p, p\rangle > 0$. **Answer: $\frac{41}{60}$**

Method 2 ⇝ Properties of Inner Product and Norm

Examples of Method-2: Properties of Inner Product and Norm

A	1	Let $V = \mathbb{R}^2$ with inner product defined by $\langle u, v\rangle = u_1v_1 + 3u_2v_2$. Let $u = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$. **1.** Find the length of u and v. **2.** Find distance between u and v. **3.** Find angle between u and v. **4.** Verify triangle inequality. **5.** Verify Cauchy-Schwartz inequality. **Answer:** $(4, 7)$, $\sqrt{109}$, $\pi - \cos^{-1}\left(\frac{22}{28}\right)$
A	2	Let $V = \mathbb{R}^2$ with inner product defined by $\langle u, v\rangle = 3u_1v_1 + 2u_2v_2$. Let $u = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$. **1.** Find the length of u and v. **2.** Find distance between u and v. **3.** Find angle between u and v. **4.** Verify triangle inequality. **5.** Verify Cauchy-Schwartz inequality. **Answer:** $(\sqrt{20}, \sqrt{35})$, $\sqrt{75}$, $\pi - \cos^{-1}\left(\frac{10}{\sqrt{20}\sqrt{35}}\right)$
A	3	Let $V = \mathbb{P}_2$ with $p = a_0 + a_1x + a_2x^2$ and $q = b_0 + b_1x + b_2x^2$ and inner product defined by $\langle p, q\rangle = a_0b_0 + a_1b_1 + a_2b_2$. Let $p = x - x^2$ and $q = 7 + 3x + 3x^2$. **1.** Find the length of p and q. **2.** Find distance between p and q. **3.** Find angle between p and q. **4.** Verify triangle inequality. **5.** Verify Cauchy-Schwartz inequality. **Answer:** $(\sqrt{2}, \sqrt{67})$, $\sqrt{69}$, $\frac{\pi}{2}$

A	4	For $A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$ and $B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$ the inner product on $\mathbb{M}_{22}$ be defined as $\langle A, B\rangle = a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2$. Let $A = \begin{bmatrix} 2 & 6 \\ 1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$ then verify Cauchy-Schwarz inequality and find the angle between A and B. **Answer:** $\boldsymbol{\theta = \cos^{-1}\left(\frac{19}{10\sqrt{7}}\right)}$
B	5	Find an angle between t and sin t for V is an inner product space of all continuous function on $[0, \pi]$ with the inner product defined by $\langle f, g\rangle = \int_0^{\pi} f(t)\, g(t)\, dt$. **Answer:** $\boldsymbol{\theta = 0.68}$
B	6	Compute d(f, g) and $\|f\|$ for $\langle f, g\rangle = \int_0^1 f(x)\, g(x)\, dx$, where $f(x) = x$ and $g(x) = e^x$ in set of all continuous functions in [0,1]. i. e., C[0,1] **Answer:** $\boldsymbol{\frac{1}{\sqrt{3}}}$, $\boldsymbol{\sqrt{\frac{e^2}{2} - \frac{13}{6}}}$
A	7	If $u = (1, 3, -2, 7), v = (0, 7, 2, 2)$ & $w = (1, -1, 0, 3)$ are vectors in the Euclidean space $\mathbb{R}^4$ then find d(u, v), d(v, w) and d(w, u). Also verify triangle inequality (Minkowski's inequality). **Answer:** $\boldsymbol{\sqrt{58}}$, $\boldsymbol{\sqrt{70}}$, $\boldsymbol{6}$
C	8	Let V be an inner product space. For $u, v \in V$ prove that $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$ (**Parallelogram Law**).
A	9	Find $u \cdot v$ given that $\|u + v\| = 1$ and $\|u - v\| = 5$ in an inner product space. **Answer:** $\boldsymbol{-6}$

Method 3 ⇝ Orthogonal and Orthonormal Sets

Examples of Method-3: Orthogonal and Orthonormal Vectors

A	1	Let $u = (1, 1, -2)$ and $v = (a, -1, 2)$. For what values of 'a', u and v orthogonal with respect to the Euclidean inner product. **Answer: 5**
A	2	Show that $\{(0, 1, 0, -1), (1, 0, 1, 0), (-1, 1, 1, 1)\}$ is orthogonal with respect to the Euclidean inner product on $\mathbb{R}^4$.
A	3	Let $\mathbb{R}^3$ have the Euclidean inner product. For which value of k, u and v are orthogonal? **1.** u = (k, k, 1) and v = (k, 5, 6). **2.** u = (2, 1, 3) and v = (1, 7, k). **Answer: (1) $k = -2$ & -3, (2) $k = -3$**
A	4	If u and v are orthogonal unit vectors, then what is the distance between u and v. Justify your answer. **Answer: $d(u, v) = \|u - v\| = \sqrt{2}$**
C	5	Find all vectors in $\mathbb{R}^3$ of Euclidean norm 1 that are orthogonal to u = (1, 1, 1) and v = (1, 1, 0). **Answer: $\pm \frac{1}{\sqrt{2}}(-1, 1, 0)$**
B	6	Verify Pythagorean theorem for the vectors u = (3, 0, 1, 0, 4, −1) and v = (−2, 5, 0, 2, −3, −18). **Answer: $\|u + v\|^2 = \|u\|^2 + \|v\|^2 = 393$**
A	7	Determine whether the given set of vectors is orthogonal with respect to the Euclidean inner product. $\{(1, -1, 2), (0, 2, -1), (-1, 1, 1)\}$. **Answer: No**
A	8	Show that $V = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthonormal set in $\mathbb{R}^3$ with Euclidean inner product.

B	9	Show that given set of vectors is orthogonal with respect to the Euclidean inner product on $\mathbb{R}^3$ and convert it into an orthonormal set by normalizing the vectors. $\left\{\left(\frac{1}{5},\frac{1}{5},\frac{1}{5}\right),\left(-\frac{1}{2},\frac{1}{2},0\right),\left(\frac{1}{3},\frac{1}{3},-\frac{2}{3}\right)\right\}$. **Answer:** $\mathbf{w_1}=\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$, $\mathbf{w_2}=\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0\right)$, $\mathbf{w_3}=\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},-\frac{2}{\sqrt{6}}\right)$
B	10	Verify that the vectors $u_1=(1,-1,2,-1)$, $u_2=(-2,2,3,2)$, $u_3=(1,2,0,-1)$, $u_4=(1,0,0,1)$ form an orthogonal basis for $\mathbb{R}^4$ with the Euclidean inner product. Also convert to an orthonormal set by normalizing the vectors. **Answer:** $\mathbf{w_1}=\left(\frac{1}{\sqrt{7}},\frac{-1}{\sqrt{7}},\frac{2}{\sqrt{7}},\frac{-1}{\sqrt{7}}\right)$, $\mathbf{w_2}=\left(\frac{-2}{\sqrt{21}},\frac{2}{\sqrt{21}},\frac{3}{\sqrt{21}},\frac{2}{\sqrt{21}}\right)$, $\mathbf{w_3}=\left(\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}},0,\frac{-1}{\sqrt{6}}\right)$, $\mathbf{w_4}=\left(\frac{1}{\sqrt{2}},0,0,\frac{1}{\sqrt{2}}\right)$
C	11	Find an orthogonal basis for $\mathbb{R}^3$ containing the vectors $u=(3,5,1)$, $v=(2,-2,4)$ using the Euclidean inner product and hence convert in orthonormal basis. **Answer:** $\mathbf{w}=(-11,5,8)$ **and** $\left\{\frac{\mathbf{u}}{\sqrt{35}},\frac{\mathbf{v}}{\sqrt{24}},\frac{\mathbf{w}}{\sqrt{210}}\right\}$

Method 4 ⇝ Gram-Schmidt Orthogonalization

Examples of Method-4: Gram-Schmidt Orthogonalization

A	1	Find the orthogonal projection of $u = (1, -2, 3)$ along $v = (1, 2, 1)$ in $\mathbb{R}^3$ with respect to the Euclidean inner product. **Answer: 0**
A	2	Find the projection of $u = (1, -2, 3)$ along $v = (2, 5, 4)$ in $\mathbb{R}^3$. **Answer:** $\left(\frac{8}{45}, \frac{20}{45}, \frac{16}{45}\right)$
A	3	Let $u = \begin{pmatrix}7\\6\end{pmatrix}, v = \begin{pmatrix}4\\2\end{pmatrix}$, then find $\text{proj}_v u$, $u - \text{proj}_v u$, also verify that v & $\text{proj}_v u$ are orthogonal to $u - \text{proj}_v u$. **Answer:** $\begin{pmatrix}8\\4\end{pmatrix}$ **and** $\begin{pmatrix}-1\\2\end{pmatrix}$
A	4	Let $\mathbb{R}^2$ have the Euclidean inner product. Use Gram Schmidt process to transform the basis vectors $u_1 = (2,3), u_2 = (4,1)$ into an orthonormal basis. **Answer:** $\left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right), \left(\frac{3}{\sqrt{13}}, -\frac{2}{\sqrt{13}}\right)$
A	5	Let $\mathbb{R}^2$ have the Euclidean inner product. Use Gram Schmidt process to transform the basis vectors $u_1 = (2,1), u_2 = (1,2)$ into an orthonormal basis. **Answer:** $\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right), \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$
B	6	Let B be the basis for $\mathbb{R}^3$ given by $B = \{(1,1,1), (-1,1,0), (-1,0,1)\}$. Apply the Gram – Schmidt process on B to find an orthonormal basis for $\mathbb{R}^3$. **Answer:** $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)$

B	7	Let $\mathbb{R}^3$ have Euclidean inner product. Use Gram-Schmidt process to transform the basis vectors $\bar{u}$, $\bar{v}$, $\bar{w}$ into an orthonormal basis, where $u = (2,1,3), v = (1,2,3), w = (1,1,1)$. **Answer:** $\frac{1}{\sqrt{14}}(2,1,3),\ \frac{1}{3\sqrt{42}}(-12,15,3),\ \frac{1}{\sqrt{3}}(1,1,-1)$
B	8	Consider the vector space $\mathbb{R}^3$ with Euclidean inner product. Apply the Gram-Schmidt process to transform the basis vectors $u_1 = (1,1,1), u_2 = (0,1,1), u_3 = (0,0,1)$ into an orthogonal basis $\{v_1, v_2, v_3\}$ then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{q_1, q_2, q_3\}$. **Answer:** $v_1 = (1,1,1),\ v_2 = (-2,1,1),\ v_3 = (0,-1,1)$ $q_1 = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right),\ q_2 = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right),$ $q_3 = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
C	9	Let $\mathbb{R}^3$ have the inner product defined by $\langle (x_1,x_2,x_3), (y_1,y_2,y_3)\rangle = x_1y_1 + 2x_2y_2 + 3x_3y_3$. Apply the Gram-Schmidt process to transform the vectors $(1,1,1), (1,1,0)$ and $(1,0,0)$ into orthonormal vectors. **Answer: Orthogonal:** $(1,1,1),\ \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right),\ \left(\frac{2}{3}, -\frac{1}{3}, 0\right)$ **Orthonormal:** $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}\right), \left(\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, 0\right)$
C	10	Let U be the subspace of $\mathbb{R}^4$ with the basis $B = \left\{\begin{bmatrix}-1\\1\\1\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\\1\end{bmatrix}\right\}$ where the inner product is the dot product. Find an orthonormal basis using Gram Schmidt process. **Answer:** $\frac{1}{\sqrt{3}}\begin{bmatrix}-1\\1\\1\\0\end{bmatrix},\ \frac{1}{\sqrt{6}}\begin{bmatrix}-1\\-2\\1\\0\end{bmatrix},\ \frac{1}{\sqrt{6}}\begin{bmatrix}1\\0\\1\\2\end{bmatrix}$

Method 5 ⇝ QR Decomposition

Examples of Method-5: Orthogonal Matrices and QR Decomposition

B	1	Find the QR decomposition of $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$ & $\begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{\sqrt{6}}{2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}$
B	2	Find the QR decomposition of $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$ & $\begin{bmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} & -\frac{1}{\sqrt{3}} \\ 0 & 0 & \frac{2\sqrt{6}}{3} \end{bmatrix}$
A	3	Find the QR decomposition of $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and verify that $QR = A$. **Answer:** $\begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$ & $\begin{bmatrix} \sqrt{5} & \sqrt{5} \\ 0 & \sqrt{5} \end{bmatrix}$

A	4	Find the QR decomposition of $A = \begin{bmatrix} 1 & 1 \\ -2 & 1 \\ 2 & 1 \end{bmatrix}$ and verify that $QR = A$. **Answer:** $\begin{bmatrix} \frac{1}{3} & \frac{8}{3\sqrt{26}} \\ -\frac{2}{3} & \frac{11}{3\sqrt{26}} \\ \frac{2}{3} & \frac{7}{3\sqrt{26}} \end{bmatrix}$ & $\begin{bmatrix} 3 & \frac{1}{3} \\ 0 & \frac{\sqrt{26}}{3} \end{bmatrix}$
C	5	Find the QR decomposition of $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 1 & 0 \\ -1 & 1 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$ & $\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$
C	6	Find the QR decomposition of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \frac{1}{2} & -\frac{3}{\sqrt{12}} & 0 \\ \frac{1}{2} & \frac{1}{\sqrt{12}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{2} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{6}} \\ \frac{1}{2} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{6}} \end{bmatrix}$ & $\begin{bmatrix} 2 & \frac{3}{2} & 1 \\ 0 & \frac{3}{\sqrt{12}} & \frac{2}{\sqrt{12}} \\ 0 & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}$

Method 6 ⇝ Least Squares Solutions

Examples of Method-6: Least Squares Solutions

A	1	Find the least square solution for system $4x_1 - 3x_2 = 12$, $2x_1 + 5x_2 = 32$, $3x_1 + x_2 = 21$. **Answer:** $\begin{bmatrix} \mathbf{230/39} \\ \mathbf{155/39} \end{bmatrix}$
A	2	Find a least square solution of the inconsistent system $AX = B$ for $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \mathbf{1} \\ \mathbf{2} \end{bmatrix}$
B	3	Find the least square solution of the linear system $AX = B$ given by $2x - 2y = 2$, $x + y = -1$, $3x + y = 1$. Also find orthogonal projection of B on the column space of A. **Answer:** $\begin{bmatrix} \mathbf{3/7} \\ \mathbf{-2/3} \end{bmatrix}$, $\begin{bmatrix} \mathbf{46/21} \\ \mathbf{-5/21} \\ \mathbf{13/21} \end{bmatrix}$
B	4	Find the least square solution of $AX = B$ given by $x_1 + x_2 = 7$, $-x_1 + x_2 = 0$, $-x_1 + 2x_2 = -7$ & find the orthogonal projection of B onto the column space of A. **Answer:** $\begin{bmatrix} \mathbf{5} \\ \mathbf{1/2} \end{bmatrix}$, $\begin{bmatrix} \mathbf{11/2} \\ \mathbf{-9/2} \\ \mathbf{-4} \end{bmatrix}$

B	5	Find the orthogonal projection of $\bar{u} = (2, 1, 3)$ on the subspace of $\mathbb{R}^3$ spanned by the vectors $v_1 = (1, 1, 0), v_2 = (1, 2, 1)$. **Answer**: $\begin{bmatrix} 2/3 \\ 7/3 \\ 5/3 \end{bmatrix}$
C	6	Find the orthogonal projection of $u = (6, 3, 9, 6)$ on the subspace of $\mathbb{R}^4$ spanned by the vectors $v_1 = (2, 1, 1, 1), v_2 = (1, 0, 1, 1), v_3 = (-2, -1, 0, -1)$. **Answer**: $\begin{bmatrix} 7 \\ 2 \\ 9 \\ 5 \end{bmatrix}$